Peer Response

Dear Panigiotis,

Your insightful post highlights several critical challenges to the large-scale adoption of Multi-Agent Systems (MAS) in manufacturing. The distributed nature of MAS creates concerns related to coordination, interoperability, and emergent behavior. For example, autonomous agents may optimize local objectives while unintentionally generating suboptimal outcomes at the system level.

Fortunately, recent research on agentic AI has proposed several strategies to address these challenges. The primary approaches for aligning agent objectives with human intentions include inverse reinforcement learning and preference learning. Meanwhile, safety assurance can be enhanced through adversarial training, formal verification, and rigorous testing protocols (Biswas and Sarkar, 2026). However, despite these promising developments, there remains a lack of mature governance frameworks specifically designed for MAS architectures in manufacturing environments.

I strongly agree with your emphasis on human oversight as a means of maintaining organizational control over decentralized agents. A practical approach is to position human operators as supervisors of the decision-making loop, allowing agent autonomy to adapt according to situational requirements. For instance, agents could operate with greater autonomy during routine scenarios, while human monitoring and intervention would increase during exceptional or high-risk situations (Suzanne Barber, Goel and Martin, 2000).

I also particularly agree with your recommendation for a hybrid architecture that combines decentralized agents with elements of centralized control. Several recent frameworks follow the principle of centralized governance with decentralized execution. For example, Farahani, Khan and Wuest (2026) proposes a hybrid multi-agent framework for prescriptive maintenance in which a Large Language Model (LLM)-based agent functions as a central orchestrator, coordinating multiple Small Language Model (SLM)-based agents responsible for domain-specific tasks. Although SLM agents possess more limited reasoning capabilities, they can make context-specific decisions efficiently, reducing dependence on centralized resources while preserving overall system coordination.

To conclude, hybrid architectures that combine decentralized decision-making with centralized coordination offer one of the most promising paths toward scalable and trustworthy implementation.

REFERENCES

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